

VenkateshVemala

AI Engineer

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Profile

AI Engineer with hands-on experience in NLP, LLMs and healthcare analytics. Skilled in fine-tuning transformer models like BioBERT for radiology exam classification, automating data-quality insights, and building analytical dashboards using Qlik and Power BI. Previously a GCP Data Engineering Associate at Accenture, bringing strong foundations in data manipulation, preprocessing, and reporting to support end-to-end AI and analytics workflows.

Professional Experience

AI Engineer/Data Analyst, Siemens Healthineers 01/2024 – Present
AI Engineering (Agentic AI Focus) Erlangen, Germany

- Enhanced BioBERT for precise radiology body-part classification in teamplay insights.
- Developed logic for mismatch detection and derived data-quality insights from outputs. Achieved a 87% accuracy.
- Applied early agentic AI patterns by modularizing tasks such as validation, consistency checks, and insight generation into semi-autonomous processing steps.
- Investigated LLM and NLP techniques for automating exam description interpretation.
- Documented findings, model behavior, and performance metrics for internal teams and stakeholders.

Data Analytics

- Built Qlik Sense dashboards to visualize KPIs, mismatch trends, and scanner-level performance insights in teamplay.
- Collaborated with stakeholders to translate requirements into effective analytical dashboards.
- Performed data cleaning, transformation, and modeling to ensure accurate reporting across teams.

Data Engineering Associate, Accenture 09/2022 – 10/2023
Bengaluru, India

- Manipulated and transformed datasets in GCP (BigQuery, Dataflow, GCS) based on team requirements.
- Wrote SQL queries and transformation logic for reporting needs.
- Worked closely with senior data engineers to implement assigned data-processing tasks.
- Validated processed data, resolved minor quality issues, and ensured accuracy in delivered datasets.

Education

M.Sc. Data Science, Friedrich Alexander University 10/2023 – Present
Specialization in Artificial Intelligence/Machine Learning Erlangen, Germany

B.E. Computer Science, Bharath University 08/2018 – 05/2022
Specialization in Software Development, System Design Chennai, India

Technical Skills

- Machine Learning Algorithms
 - Gen AI
 - Image classification, segmentation, and feature extraction
 - Google Cloud Platform
 - BioBERT, BERT-based models
 - NLP, LangChain, LLM/RAG Data Pipelines
 - PyTorch
- Agentic AI
 - Deep Learning
 - Python
 - Qlik Sense
 - ML pipelines & experimentation
 - SQL & Query Optimization

Projects

Brain Tumor Classification System (Computer Vision, Deep Learning, Flask)

- Built a CNN-based classification model using VGG16 to detect four types of brain tumors (Glioma, Meningioma, Pituitary, and No Tumor) from MRI images.
- Preprocessed and augmented MRI datasets, trained the model using TensorFlow/Keras, and achieved high prediction accuracy.
- Deployed the model as a web application using Flask, enabling users to upload MRI images and receive real-time tumor predictions with confidence scores.

Certificates

- Google Cloud Certified Associate Cloud Engineer
- Understanding Agentic AI Agentic AI

Soft Skills

- Leadership & Critical Thinking
 - Attention to Domain details
- Continuous Learning
 - Teamwork in cross-functional settings

Languages

English

C1

German

A1